

## ACAPULCO INTL, Mexico

WMO: 768056

Lat: 16.7628N

Lon: 99.7494W

Elev: 10

StdP: 14.69

Time Zone: -6.00 (MEX)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 2             | 66.6       | 69.4 | 56.8                          | 68.7 | 74.2 | 58.9 | 74.4 | 74.7 | 18.4                  | 84.0 | 16.9 | 84.1 | 2.3                      | 0    | 0.350 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.2                   | 91.8            | 79.9 | 91.4 | 79.7 | 90.0 | 78.8 | 82.2                | 89.6 | 81.4 | 88.6 | 80.7 | 87.7 | 9.7                  | 240  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 80.5                            | 158.2 | 87.9 | 79.1 | 151.3 | 86.6 | 78.9 | 149.9 | 86.4 | 46.1          | 90.1 | 45.2 | 89.4 | 44.6 | 87.9 | 85.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |
| 18.0              | 15.5 | 13.3 | 64.5                       | 95.0 | 3.1                | 2.6 | 62.2                                               | 96.9 | 60.4       | 98.5 | 58.6       | 99.9 | 56.3       | 101.9 |
|                   |      |      | 58.6                       | 83.9 | 2.6                | 1.2 | 56.8                                               | 84.7 | 55.3       | 85.4 | 53.8       | 86.1 | 52.0       | 87.0  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 81.1   | 79.0  | 78.6  | 78.4  | 79.3  | 82.2  | 83.2  | 83.1  | 83.1  | 82.1  | 82.2  | 81.5  | 79.9  |
|                                                                        | DBStd       | 2.83   | 2.14  | 2.11  | 2.51  | 2.56  | 2.54  | 2.15  | 2.01  | 2.15  | 2.11  | 1.98  | 1.87  | 2.06  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD50       | 11339  | 899   | 801   | 881   | 879   | 1000  | 996   | 1027  | 1025  | 963   | 997   | 944   | 927   |
|                                                                        | CDD65       | 5864   | 434   | 381   | 416   | 429   | 535   | 546   | 562   | 560   | 513   | 532   | 494   | 462   |
|                                                                        | CDH74       | 62264  | 3787  | 3414  | 3881  | 4289  | 6218  | 6530  | 6778  | 6615  | 5590  | 5877  | 5097  | 4188  |
|                                                                        | CDH80       | 22574  | 1244  | 1099  | 1249  | 1431  | 2432  | 2612  | 2729  | 2633  | 1908  | 2028  | 1797  | 1412  |
| (14) Wind                                                              | WSAvg       | 5.6    | 5.0   | 5.3   | 5.8   | 6.1   | 6.1   | 6.0   | 5.9   | 5.9   | 5.7   | 5.3   | 4.9   | 4.7   |
| (15) Precipitation                                                     | PrecAvg     | 48.8   | 0.4   | 0.3   | 0.2   | 0.0   | 1.5   | 9.2   | 8.5   | 9.8   | 11.8  | 5.5   | 0.8   | 0.3   |
|                                                                        | PrecMax     | 69.0   | 2.6   | 5.0   | 1.7   | 0.5   | 8.1   | 17.4  | 18.7  | 21.7  | 18.9  | 16.4  | 6.3   | 1.2   |
|                                                                        | PrecMin     | 29.8   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 5.3   | 1.8   | 2.2   | 5.4   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 10.4   | 0.8   | 1.1   | 0.5   | 0.1   | 2.0   | 3.6   | 4.3   | 5.1   | 3.9   | 4.1   | 1.3   | 0.4   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 91.7  | 91.1  | 91.2  | 90.0  | 91.8  | 93.2  | 93.0  | 93.4  | 91.7  | 91.7  | 91.7  | 91.5  |
|                                                                        |             | MCWB   | 76.1  | 77.0  | 78.2  | 76.6  | 79.8  | 80.7  | 81.2  | 82.3  | 80.3  | 80.5  | 79.6  | 78.9  |
|                                                                        | 2%          | DB     | 89.3  | 88.1  | 88.2  | 89.2  | 90.0  | 91.4  | 91.3  | 91.6  | 89.9  | 89.9  | 91.0  | 89.5  |
|                                                                        |             | MCWB   | 76.1  | 75.7  | 75.8  | 76.4  | 78.3  | 79.5  | 80.1  | 80.6  | 79.5  | 79.7  | 79.5  | 77.5  |
|                                                                        | 5%          | DB     | 87.6  | 86.7  | 87.5  | 87.9  | 89.6  | 89.8  | 89.8  | 89.9  | 89.2  | 88.2  | 89.2  | 87.9  |
|                                                                        |             | MCWB   | 75.2  | 75.3  | 75.4  | 76.1  | 78.0  | 78.7  | 79.2  | 79.5  | 79.2  | 79.0  | 78.2  | 76.5  |
|                                                                        | 10%         | DB     | 85.8  | 85.8  | 85.8  | 86.3  | 88.2  | 89.3  | 89.2  | 89.3  | 87.8  | 87.8  | 87.7  | 86.2  |
|                                                                        |             | MCWB   | 74.9  | 75.1  | 75.1  | 75.8  | 77.6  | 78.6  | 79.0  | 79.2  | 78.9  | 78.9  | 77.7  | 76.3  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 79.8  | 79.8  | 79.9  | 79.9  | 82.0  | 82.7  | 82.6  | 83.7  | 81.9  | 82.5  | 82.2  | 80.8  |
|                                                                        |             | MCDB   | 88.0  | 88.0  | 88.3  | 87.0  | 89.7  | 90.0  | 90.0  | 91.7  | 89.6  | 88.7  | 89.8  | 89.0  |
|                                                                        | 2%          | WB     | 78.2  | 78.2  | 78.4  | 78.8  | 80.7  | 81.5  | 81.5  | 81.9  | 80.8  | 81.0  | 80.7  | 79.4  |
|                                                                        |             | MCDB   | 86.4  | 86.1  | 86.0  | 86.2  | 87.8  | 88.6  | 88.6  | 89.5  | 88.0  | 87.2  | 87.3  | 86.8  |
|                                                                        | 5%          | WB     | 77.2  | 77.1  | 77.3  | 77.7  | 79.8  | 80.5  | 80.6  | 80.9  | 79.9  | 80.3  | 79.8  | 78.3  |
|                                                                        |             | MCDB   | 85.1  | 84.8  | 84.7  | 85.3  | 86.6  | 87.3  | 87.6  | 88.1  | 86.7  | 86.3  | 86.1  | 85.3  |
|                                                                        | 10%         | WB     | 76.1  | 76.1  | 76.3  | 76.9  | 79.1  | 79.7  | 79.9  | 80.0  | 79.2  | 79.5  | 78.9  | 77.3  |
|                                                                        |             | MCDB   | 83.5  | 83.4  | 83.5  | 84.5  | 85.7  | 86.3  | 86.8  | 86.9  | 85.6  | 85.5  | 85.2  | 84.2  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.8  | 14.8  | 15.0  | 14.5  | 12.4  | 11.0  | 11.2  | 11.3  | 10.6  | 10.7  | 12.7  | 13.9  |
|                                                                        |             | MCDBR  | 16.3  | 15.9  | 16.1  | 15.0  | 12.8  | 12.3  | 12.3  | 12.8  | 12.5  | 11.9  | 14.6  | 15.2  |
|                                                                        |             | MCWBR  | 8.8   | 9.0   | 8.9   | 8.3   | 6.6   | 6.0   | 6.8   | 7.0   | 6.4   | 6.2   | 7.4   | 8.0   |
|                                                                        | 5% WB       | MCDBR  | 14.4  | 14.4  | 14.3  | 13.9  | 12.0  | 11.3  | 11.6  | 12.1  | 11.4  | 11.1  | 12.5  | 13.7  |
|                                                                        |             | MCWBR  | 8.4   | 8.5   | 8.5   | 8.2   | 6.6   | 6.2   | 6.9   | 7.2   | 6.3   | 6.3   | 6.8   | 7.6   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.407 | 0.406 | 0.429 | 0.484 | 0.595 | 0.479 | 0.442 | 0.444 | 0.437 | 0.424 | 0.400 | 0.388 |
|                                                                        | taud        |        | 2.412 | 2.406 | 2.334 | 2.170 | 1.909 | 2.273 | 2.398 | 2.379 | 2.383 | 2.435 | 2.484 | 2.499 |
|                                                                        | Ebn at Noon |        | 279   | 285   | 281   | 264   | 233   | 259   | 270   | 272   | 275   | 276   | 279   | 282   |
|                                                                        | Edh at Noon |        | 36    | 37    | 41    | 49    | 63    | 43    | 38    | 39    | 39    | 36    | 33    | 32    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1652  | 1931  | 2163  | 2272  | 2052  | 1840  | 1971  | 1888  | 1696  | 1706  | 1679  | 1618  |
|                                                                        | RadStd      |        | 97    | 99    | 81    | 49    | 125   | 128   | 121   | 109   | 126   | 111   | 88    | 66    |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | -1.08   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page